Group Reflection – NewsBot Intelligence System

Course: ITAI 2373 – Natural Language Processing
Project: NewsBot Intelligence System – Midterm Project
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Introduction

For our midterm project, we developed a NewsBot Intelligence System using Natural Language Processing (NLP) techniques to analyze and classify news articles. We used the BBC News Classification dataset and divided the project into eight modules, with each group member responsible for two. Our goal was to understand how text preprocessing, feature extraction, grammatical analysis, sentiment analysis, machine learning, and named entity recognition work together in a complete NLP system.

Individual Contributions and Learning

Oman – Modules 1 and 2: Business Understanding and Text Preprocessing

Oman worked on understanding the business problem, exploring the dataset, and preparing news articles for analysis. In Module 1, he examined the dataset structure, categories, and article distribution to understand how NewsBot could organize news automatically. In Module 2, he applied text cleaning, tokenization, stopword removal, and lemmatization. He learned that preprocessing improves text quality and prepares articles for machine learning. One challenge was removing unnecessary information without losing meaningful words.

Tobias – Modules 3 and 4: TF-IDF and Part-of-Speech Tagging

Tobias worked on TF-IDF feature extraction and Part-of-Speech (POS) tagging. In Module 3, he learned how TF-IDF converts text into numerical features and identifies important words in news articles. In Module 4, he explored grammatical roles such as nouns, verbs, and adjectives. He learned that NLP systems analyze both word importance and sentence structure. His main challenge was understanding large TF-IDF feature sets and how grammatical tags change depending on context.

Cris – Modules 5 and 6: Dependency Parsing and Sentiment Analysis

Cris worked on dependency parsing and sentiment analysis. In Module 5, she explored grammatical relationships between words to understand sentence structure and identify important connections. In Module 6, she analyzed positive, negative, and neutral language in news articles. She learned that dependency parsing provides context, while sentiment analysis helps identify emotional tone. One challenge was interpreting articles containing factual information or mixed emotions, because automated sentiment predictions are not always accurate.

Mahdi Mohammadkhani – Modules 7 and 8: Text Classification and Named Entity Recognition

Mahdi worked on machine learning classification and Named Entity Recognition (NER). In Module 7, he trained and evaluated classification models to categorize articles into Business, Entertainment, Politics, Sport, and Tech. He learned how model predictions and evaluation metrics measure classification performance. In Module 8, he identified entities such as people, organizations, and locations. One challenge was understanding why classification models sometimes make incorrect predictions when articles contain overlapping topics or ambiguous language.

Challenges and Problem-Solving

One of our main challenges was integrating eight different NLP modules into a single working system. Each module required different techniques and produced different outputs. We needed to maintain a consistent dataset and understand how each module supported the next. For example, preprocessing prepared articles for TF-IDF, while numerical features supported machine learning classification.

We addressed these challenges by dividing responsibilities, reviewing each module's purpose, and testing the notebook to identify errors. We also learned that classification, sentiment analysis, and entity recognition can produce incorrect results, making evaluation an important part of NLP development.

Real-World Applications and Ethical Considerations

NewsBot demonstrates how NLP can help news organizations automatically categorize articles, businesses monitor company mentions and public sentiment, and researchers analyze large collections of news. These applications can save time and make information easier to organize.

However, we also recognized important ethical concerns, including biased training data, incorrect classifications, and misleading sentiment predictions. Automated results should be evaluated carefully, and human review remains important when interpreting sensitive or complicated information.

Final Reflection and Conclusion

Overall, this project improved our understanding of Python, machine learning, and Natural Language Processing. Dividing the eight modules among four members helped us focus on individual responsibilities while understanding how each technique contributes to the complete system.

Our most important lesson was that building an NLP application involves more than writing code. It requires understanding the problem, preparing data, analyzing language, evaluating results, and considering real-world limitations. NewsBot provided valuable experience in teamwork and demonstrated how different NLP techniques can work together to solve a practical problem.